



**SPE - Cross-Platform Mobile
Development**

T2-144

Monday, 14 July 2025
2:00 PM to 5:00 PM

T2-144

Monday, 14 July 2025
2:00 PM to 5:00 PM

Index number

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

QUESTIONS and ANSWERS BOOKLET

TRADE (s) : SOFTWARE PROGRAMMING AND EMBEDDED SYSTEMS (SPE)
SUBJECT: CROSS-PLATFORM MOBILE DEVELOPMENT
DURATION: 3 Hours

SUBJECT: CROSS-PLATFORM MOBILE DEVELOPMENT

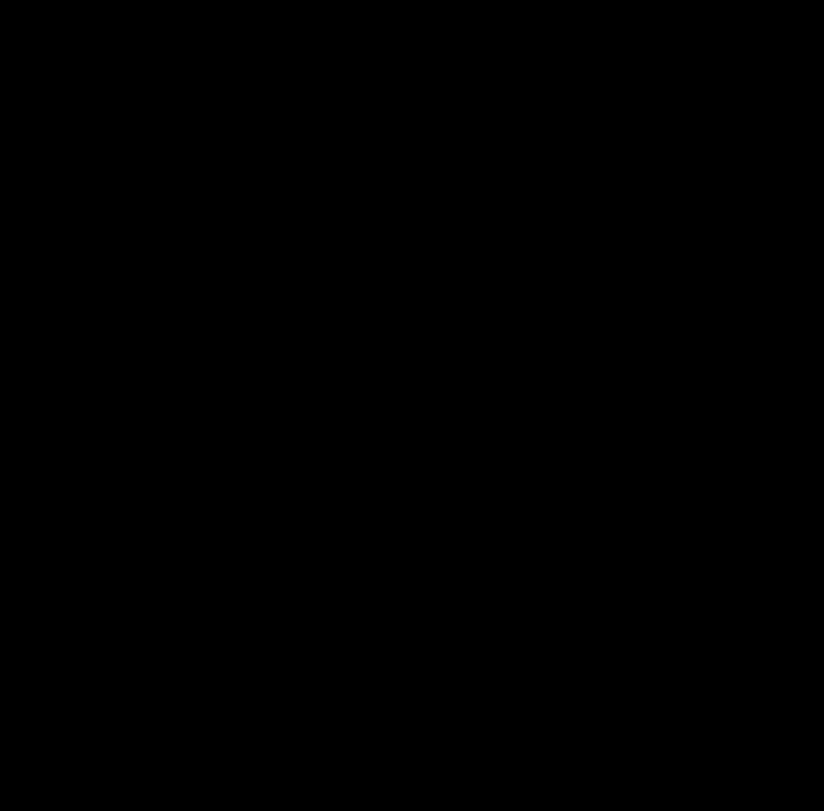
DURATION: 3 Hours

Read carefully the instructions on page 3

TOTAL MARKS...../ 100

FOR EXAMINER'S USE ONLY

[illegible]



TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

TRADE: SOFTWARE PROGRAMMING AND EMBEDDED SYSTEMS (SPE)

SUBJECT: CROSS-PLATFORM MOBILE DEVELOPMENT

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

1. This Exam paper is composed of **three sections (A, B and C)**. Follow the instructions given below and answer the indicated questions for a total of **100 marks**.

Section **A**: Fourteen (14) questions which are **Compulsory** **55 marks**

Section **B**: Among the five (5) questions, attempt any three (3) **30 marks**

Section **C**: Among the two (2) questions, attempt anyone (1) **15 marks**
2. Fill in your actual **names** and **Index number** on the provided space (Cover).
3. Do not remove or tear any page or part of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
 - Blue or Black pen.
 - Geometrical Instruments
 - Silent non-programmable calculators

SECTION A: Attempt all questions

(55 marks)

- 01.** In React native, an API is a way your mobile app talks to a backend server or third-party service to get or send data. Fill the gap in the following statements accordingly. **(4 Marks)**
- a) The _____ API is used to create animations in React Native.
 - b) In React native, _____ is a popular third-party library used to make API calls easier and more powerful.
 - c) In API requests, the _____ property is used to define whether the request is GET, POST, PUT, or DELETE.
 - d) A JSON object is usually parsed using _____ method after receiving a response.
- 02.** FlatList is a scrollable list component that displays many items efficiently by only rendering the items that are currently visible on screen. Complete the following statements with a suitable word. **(4 Marks)**
- a) _____ is a prop in FlatList that triggers a data refresh.
 - b) In FlatList, to add a separator between list items, you can use the _____ prop.
 - c) FlatList's _____ prop allows specifying how items to render initially.
 - d) To add a footer at the end of the list, FlatList uses the _____ prop.
- 03.** A library in React Native is a collection of pre-written code that helps developers build apps faster without having to write everything from scratch. Referring to that respond to the following statements using **true(T)** or **false(F)** in the provided space. **(2 Marks)**
- a) To detect network status, React Native uses the NetInfo library.....
 - b) AsyncStorage is a library used to perform image loading in React Native
 - c) React Navigation is a build-in part of React Native for managing screens and navigation.
 - d) Lottie library is used for handling complex animations in React native apps.

T2-144_Cross-Platform Mobile Development

- 04.** Designing a 3D model means creating the shape, size, and appearance of an object inside a virtual 3D space using 3D modeling software. **(4 Marks)**

Match Types of 3D Models with Their Characteristics:

List 1	List 2	Answer
a) Solids	1.Preparing model for animation by adding bones	1 =
b) Meshes	2.Sets of discrete data points in 3D space	2 =
c) Point Clouds	3.Networks of vertices and faces	3 =
d) Rigging	4.Fully enclosed and volumetric shapes	4 =

- 05.** List any 2 transformations you may know used while modeling an object using blender software. **(3 Marks)**

.....
.....

- 06.** 3D printing is a manufacturing process where a physical object is created from a digital design by building layer by layer. Circle the letter corresponding to the correct answer to each of the following sub-questions. **(4 Marks)**

- a) In FDM (Fused Deposition Modeling) technology, which material is commonly used?
- i) Photopolymer resin
 - ii) Nylon powder
 - iii) PLA filament
 - iv) UV curable
- b) In SLA (Stereo-lithography) printing, which of the following statements are true?
- i) It uses UV light to cure photopolymer resin.
 - ii) It builds objects layer-by-layer from metal powder.
 - iii) It can produce very high detail, smooth surface finishes.
 - iv) SLA printers can use flexible resins to make rubber-like parts.
 - v) It is typically more durable than FDM prints for mechanical parts.

T2-144_Cross-Platform Mobile Development

- 07.** Managing user authentication means controlling how users prove their identity when they access a system or an application. Match the items in Column A with the correct options from Column B related to the above description. **(4 Marks)**

Column A	Column B	Answer
1. SMS OTP services	A. Help recover	1 =
2. Authenticator apps	B. Unsafe security practice	2 =
3. Backup recovery codes	C. Common 2FA method	3 =
4. Sending code over unsecured email	D. Generate time-based codes	4 =

- 08.** A design team is tasked with creating a conceptual 3D model of eco-house for an international exhibition. Before beginning detailed 3D CAD modeling, the team must sketch the concept, collect relevant data, interpret the design message, dimension key elements, select appropriate measuring tools, and prepare documentation for submission. **(4 Marks)**

In the process of dimensioning various components of the eco-house 3D model, which practices are appropriate? (Circle all that apply)

- a) Indicate key measurements such as wall thickness, door height, and window dimensions.
- b) Disregard dimensioning since it is only initial concept model.
- c) Maintain consistent units throughout the drawing (meters or millimeters).
- d) Provide dimensions only for the exterior structure, omitting interior details.
- e) Label all dimensions clearly on the sketches.
- f) Use rough estimates without verifying measurements.
- g) Ensure that scaled dimensions match the drawing proportions.

T2-144_Cross-Platform Mobile Development

- 09.** 3D rendering is the process of creating a realistic or stylized image from a 3D model using computer software. **(4 Marks)**

Match rendering types in the list 1 with their application in the list 2:

List 1	Answer	List 2
A. 2D Rendering	1 =	1.Produces flat shaded images
B. 3D Rendering	2 =	2.Produces character-based output
C. Avatar Rendering	3 =	3.Produces depth, lighting, and shadow
D. Concept Sketching	4 =	4.Represents the original design idea visually

- 10.** A design team has completed the 3D modeling of smart device enclosure. Before printing, the team must prepare the model using slicing software, configure the printer, and complete post-processing to achieve a high-quality final product. Answer the following sub-questions by circling best options related to the above scenario. **(5 Marks)**

a) Why is slicing necessary before 3D printing a model?

- i) It converts the 3D model into printable layers the 3D printer can understand.
- ii) It decorates the model with colors before printing.
- iii) It translates the design into machine-readable instructions (G-code).
- iv) It adds random textures to the model.

b) When choosing slicing software, what aspects should be understood for proper selection?

- i) Whether the slicer is compatible with the specific 3D printer being used.
- ii) Whether the slicer has features like support generation and customizable settings.
- iii) Whether the slicer is famous on social media.
- iv) Whether the slicer offers control over print quality and material.
- v) Setting the printer's nozzle and bed temperatures based on material type.
- vi) Ignoring calibration because the printer is "smart enough".

T2-144_Cross-Platform Mobile Development

- 11.** Designing a mobile app prototype means creating an early, interactive model of mobile app to show how it looks and work. Match the feature from List 1 to its tool/component in the List 2 related: **(4 Marks)**

List 1	List 2	Answer
a) Shadows	1) React Navigation	1 =
b) Offline Storage	2) AsyncStorage	2 =
c) Navigation	3) Elevation	3 =
d) Animations	4) Animated	4 =

- 12.** Integrating APIs means connecting your application to another system or service (like a database, cloud platform, third-party service, or another app) by sending requests and receiving responses through an API. **(5 Marks)**

```
import javax.ws.rs.Applicationpath;
import javax.ws.rs.core.Applicaion;
@ApplicationPath("/api")
public class ApplicationConfig extends Application
{
//.....
}
```

Referring to the above segment of code, circle the correct answer for the following questions.

- a) What is the purpose of the `@ApplicationPath("/api")` annotation in `ApplicationConfig` class?
- i) It defines the base URL path for all JAX-RS resources
 - ii) It disables all REST endpoints
 - iii) It secures the API with authentication
 - iv) It specifies the database connection URL.
- b) If you want all REST endpoints to start with `/service/` instead `/api/`, what change would make?
- i) Replace `@ApplicationPath("/api")` with `@ApplicationPath("/service")`
 - ii) Change `extends Application` to `extends Service`
 - iii) Add `@Path("/service")` inside the `Applicationconfig` class
 - iv) No change is needed, JAX-RS handles it automatically

T2-144_Cross-Platform Mobile Development

- c) Which of the following are important tasks when integrating APIs and external services? (Circle all that apply)
- i) Understanding API paradigms like RESTful API and GraphQL
 - ii) Data parsing and transformation
 - iii) Ignoring CORS-related issues
 - iv) Implementing data fetching mechanisms
 - v) Configuring external services properly
 - vi) Handling exceptions effectively

13. A developer is preparing a mobile app called LearnPro to launch on app stores. Your technical knowledge is needed to ensure proper submission, optimization, and user documentation based on real publishing requirements. Answer the following sub-questions by circling the best option. **(4 Marks)**

- a) To brand LearnPro before submission, which elements should you apply to visually represent the app during user installation and launch?
- i) Theme fronts and SDK
 - ii) App icon and splash screen
 - iii) Database Model and service endpoints
- b) While preparing LeanPro for app stores, you must upload certain assets. Which assets must you actively provide to complete the app listing?
- i) User reviews
 - ii) App name, description, screenshots, and preview videos
 - iii) Project GitHub repository
- c) If you want to avoid LearnPro's rejection during app review, what practice should you apply beforehand?
- i) Ignore store guidelines
 - ii) Familiarize yourself with app store policies and rules
 - iii) Increase app price

14. A 3D model is a mathematical representation of three-dimensional object using specialized software. (Circle all that apply to each of the following sub-questions) **(4 Marks)**

- i) You are exporting a complex 3D model from your CAD software for 3D printing. Which of the following is not a correct file format use for this purpose?
- a) .STL
 - b) .DWD
 - c) .OBJ
 - d) .STEP

T2-144_Cross-Platform Mobile Development

- ii) You need to apply meshing to 3D model for finite element analysis. Which of the following is not a correct action to take?
- a) Use the mesh generation tool to divide 3D Model into smaller elements for analysis
 - b) Apply meshing to prepare the model for simulation
 - c) Verify the mesh quality before starting the simulation
 - d) Export the model as a .JPG file analysis

Section B: Attempt any three (3) questions

(30 marks)

-
- 15.** Write a TextInput component that logs its value on change. **(10marks)**
- 16.** Create a FlatList with separators and 4 hardcoded items. **(10marks)**
- 17.** The following React Native component is intended to display a button that triggers an alert, but it has issues. Analyze the code and perform the tasks below. **(10marks)**

```
import React from 'react';
import { Button, Alert } from 'react-native';

const AlertButton = () => (
  <Button title="Show Alert" onPress={() => Alert.alert('Hello',
    'This is an alert')} />
);

export default AlertButton;
```

- a) Explain the purpose of this component and identify the main issue preventing it from working as intended.
- b) Fix the issue identified in part (a) without writing the full code.
- c) Analyze the Alert.alert call, what does it do, and how you could improve it by adding user interaction?
- d) Modify the Alert.alert call to include a "Cancel" and "OK" button, and log a message when "OK" is pressed. Write only the modified Alert.alert line.

- 18.** The following fetch code is supposed to display a post title, but it shows "Loading..." indefinitely. Analyze the code and perform the tasks below. **(10marks)**

```
import React from 'react';
import { Text } from 'react-native';

const FetchPost = () => {
  let title = '';

  fetch('https://jsonplaceholder.typicode.com/posts/1')
    .then((response) => response.json())
    .then((data) => (title = data.title));

  return <Text>{title || 'Loading...'}</Text>;
};
export default FetchPost;
```

- Explain the purpose of this component and identify the main issue causing the "Loading..." text to persist.
 - Analyze the fetch call. What does it do, and what is missing from handling errors?
 - Suggest a fix to make the component re-render with the fetched title, without writing the full code.
 - Add error handling to the fetch call by including a `.catch()` block that logs the error. Write only the modified fetch chain.
- 19.** The following AsyncStorage code fails to save and retrieve data correctly. Analyze the code and perform the tasks below. **(10marks)**

```
import React from 'react';
import AsyncStorage from '@react-native-async-storage/async-storage';

const SaveData = () => {
  AsyncStorage.setItem('name', 'John');
  const name = AsyncStorage.getItem('name');
  console.log(name);
  return null;
};
export default SaveData;
```

- Explain the purpose of this component and identify the main issue with the AsyncStorage usage.
- Analyze the `console.log(name)` line. Why does it log a Promise instead of the value "John"?
- Suggest a fix to correctly retrieve and log the value, without writing the full code.

T2-144_Cross-Platform Mobile Development

- d) Explain why `useEffect` is necessary to handle this operation in a React Native component.
- e) Write the `useEffect` block to fix the `AsyncStorage` operations, assuming an async function `saveAndLoad` is defined.

Section C: Attempt only one (1) question

(15 marks)

- 20.** Create a React Native component that uses `MapView` to display a map centered on a specific location (latitude: 37.78825, longitude: -122.4324) with a marker. **(15marks)**
- 21.** The following authentication code lacks a key feature. Analyze the code and perform the tasks below. **(15marks)**

```
import React, { useState } from 'react';
import { TextInput, Button } from 'react-native';

const LoginScreen = () => {
  const [username, setUsername] = useState('');
  const [password, setPassword] = useState('');
  const login = () => console.log('Logging in:', username, password);
  return (
    <>
      <TextInput value={username} onChangeText={setUsername}
placeholder="Username" />
      <TextInput value={password} onChangeText={setPassword}
placeholder="Password" />
      <Button title="Login" onPress={login} />
    </>
  );
};
export default LoginScreen;
```

- a) Explain the purpose of this component and how user input is handled.
- b) Analyze the login function. What security feature is missing, and how would you add it?
- c) Suggest a modification to validate the inputs before login, without writing the full code.
- d) Write the modified password `TextInput` with `secureTextEntry`.

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

[illegible]

**Do not
write in
this margin**

T2-144_Cross-Platform Mobile Development

[illegible]

**Do not
write in
this margin**

**Do not
write in
this margin**

T2-144_Cross-Platform Mobile Development

[illegible]

**Do not
write in
this margin**

**Do not
write in
this margin**

T2-144_Cross-Platform Mobile Development

[illegible]

**Do not
write in
this margin**

**Do not
write in
this margin**

T2-144_Cross-Platform Mobile Development

[illegible]

**Do not
write in
this margin**

**Do not
write in
this margin**

